# *Local Income Taxes and Residential Sorting*
## *A Unified Binding-State Decomposition*

Matthew Hockert†

† *Department of Applied Economics, University of Minnesota*

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# Question

- When a Pennsylvania municipality changes its earned-income-tax rate, do workers *sort* — relocating residence or workplace?
- Three margins of response, all observable in LODES origin-destination data:
  - **Live-and-work** workers: residents employed locally.
  - **In-commuters**: nonresidents employed in the muni.
  - **Out-commuters**: residents employed elsewhere.
- Empirical challenge: the bill is piecewise $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only *moves the bill* when its side binds.

# Roadmap

- **Question:** Do PA EIT rate changes cause residential / job sorting?
- **Data:** LODES OD (2002–2019, PA) + DCED EIT panel + AFR fiscal panel.
- **Methods:** One muni × year panel; both $\Delta\tau^{\text{res}}$ and $\Delta\tau^{\text{nr}}$ in the same regression; decomposed by direction × binding state. Five estimators.
- **This deck:** Setup → cell counts → headline results → triangulation → event studies → placebo + robustness.

# Institutional Setting (PA EIT, post-Act 32)

For a PA resident working outside Philadelphia:

$$T_{ij} = \max\left(\tau_j^{\text{res}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr}}\right).$$

Liability allocation: work muni collects up to its nonres rate; home muni + SD collect any residual.

- **Act 32 (2008 / effective 2012)** standardized employer withholding, residency certification, and consolidated tax-collection districts. Statutory rates were unchanged — the change was administrative.
- **Empirical implication.** Pre-Act-32, the same statutory rate was weakly enforced. Post-Act-32, the binding rule above actually moves the bill.
- **Sample.** 2002–2019 (avoid COVID).

# Data

- **LODES OD** (2002–2019, PA): block-level origin-destination flows aggregated to PA county subdivisions (cousubs) via TIGER 2020 block centroid. Recovers all 2,561 PA cousubs (vs 1,479 under the prior tract-centroid join).
- **DCED EIT panel**: muni × year resident, school district, and nonresident rates. Reparsed XLS to recover SD identifiers. Nonres rate has reporting gaps (handled via three sample variants).
- **DCED AFR**: muni fiscal panel for controls (share-based: police / streets / debt service as a share of total expenditures; EIT / real estate / intergov as a share of total revenues).
- **Sample**: ∼2,500 cousubs × 18 years.

# Setup: One Panel, Three Treatments

**Unit** $i$: a PA cousub. **Time** $t$: year. One row per $(i, t)$.

**Outcomes per row** (eight; covered simultaneously):

- `log_live_and_work`, `log_in_commuters`, `log_out_commuters`
- `log_total_jobs`, `log_total_residents`
- `share_live_in`, `share_nonres`, `share_work_outside`

**Three treatments per row**, each separately decomposed by cells:

- $\Delta\tau_{it}^{\text{muni}}$: cumulative muni resident-rate change.
- $\Delta\tau_{it}^{\text{SD}}$: cumulative school-district rate change.
- $\Delta\tau_{it}^{\text{nr}}$: cumulative muni nonresident-rate change.

**Per-row binding state**: $\text{home_binds}_{it} = \mathbf{1}\{\tau_{it}^{\text{home}} > \tau_{it}^{\text{nr}}\}$ where $\tau^{\text{home}} = \tau^{\text{muni}} + \tau^{\text{SD}}$. Cells classified separately for each treatment using its own first event.

# Cell Classification (per side, fixed at first event)

Each muni gets one cell label on each side, based on the binding state *just before* vs *just after* its first event.

| Cell | Direction | Binding transition |
|---|---|---|
| **Inc-bind** | Up | binding pre and post (bill ↑ by full $\Delta\tau$) |
| **Inc-becomes** | Up | flips non-binding → binding (bill rises by $\tau_{\text{post}} - \tau^{\text{nr}}$) |
| **Inc-nobind** | Up | non-binding throughout (**bill unchanged** — placebo) |
| **Dec-bind** | Down | binding pre and post (bill ↓ by full $\Delta\tau$) |
| **Dec-becomes-non** | Down | flips binding → non-binding |
| **Dec-nobind** | Down | non-binding throughout (**bill unchanged** — placebo) |
| NC | — | never-changed (control) |

Institutional rule predicts —Inc-bind— > —Inc-becomes— > —Inc-nobind— (= 0). Inc-nobind and Dec-nobind are **predicted-null placebos within the design.**

# Estimating Equation

For outcome $Y$ and estimator $E$:

$$
\begin{aligned}
Y_{it} = \alpha_i + \gamma_t &+ \sum_k \beta_k^{\text{muni}} \, \Delta\tau_{it}^{\text{muni}} \cdot \mathbf{1}\{\text{cell}_i^{\text{muni}} = k\} \\
&+ \sum_k \beta_k^{\text{SD}} \, \Delta\tau_{it}^{\text{SD}} \cdot \mathbf{1}\{\text{cell}_i^{\text{SD}} = k\} \\
&+ \sum_k \beta_k^{\text{nr}} \, \Delta\tau_{it}^{\text{nr}} \cdot \mathbf{1}\{\text{cell}_i^{\text{nr}} = k\} + \mathbf{X}_{it}'\phi + \varepsilon_{it}
\end{aligned}
$$

- All three treatments and their cell decompositions enter the *same* regression. No OD flows; no pair panel.
- **Control group:** NC (*never-changed*) munis on each side. For an NC muni, $\Delta\tau^{\text{muni}} = \Delta\tau^{\text{SD}} = \Delta\tau^{\text{nr}} = 0$, so all three cell-×-dose terms drop. NC munis identify $\gamma_t$ (the parallel time trend) and $\alpha_i$. Cell coefficients $\beta_k$ are dose-response *relative to NC's trend*. Sample: ∼2,400 muni-rate NC, ∼2,250 SD NC, ∼2,275 nonres NC.
- Reference cell in factor coding: `ref='NC'`.
- $\mathbf{X}_{it}$: **share-based controls, all time-varying** — expenditure shares of total expenditures (police, streets, debt service); revenue shares of total revenues (EIT, real estate, intergov).
- Fixed effects: muni $\alpha_i$, year $\gamma_t$. SE clustered at muni.

# Three Sample Variants for Missing Nonres Rates

DCED's nonres-rate reporting is patchy pre-Act-32 (∼55% coverage) and incomplete post (∼72–80%). All else equal, we run three samples:

| Sample | Treatment of NA `muni_nonres_eit` |
|---|---|
| **strict** | Drop muni-years with NA nonres rate. |
| **filled** | Impute `nonres = resident_eit` when NA (statute-anchored). |
| **drop** | Keep all years; drop NA at regression time per outcome. |

**Headline tables**: `drop` sample (largest N). `filled` and `strict` reported as robustness.

# Cell Counts (drop sample, unique munis)

| Treatment | Inc-bind | Inc-becomes | Inc-nobind | Dec-bind | Dec-becomes | Dec-nobind |
|---|---|---|---|---|---|---|
| Muni resident | 40 | 58 | 26 | 11 | 4 | 15 |
| School district (SD) | 75 | 197 | 14 | 4 | 10 | — |
| Muni nonresident | 0 | 0 | 171 | 0 | 5 | 109 |

- Three treatments share the same binding indicator (home_stack vs nonres) and same outcome panel.
- **Muni and SD events differ in count and timing**: SD events are more common, especially Inc-becomes (197 vs 58 muni).
- **Nonres-side binding decomposition collapses**: in PA, $\tau^{\text{nr}}$ almost never exceeds home_stack at the muni-self-pair level (Philadelphia is the rare exception, excluded). Effectively one Inc cell + two Dec cells.

# Muni Resident Rate: Stacked Continuous, Drop Sample

| | Inc-bind ($n$=40) | Inc-becomes ($n$=58) | Inc-nobind *placebo* ($n$=26) | Dec-bind ($n$=11) | Dec-becomes ($n$=4) | Dec-nobind *placebo* ($n$=15) |
|---|---|---|---|---|---|---|
| log(live_and_work) | **−0.283*** ** | **−0.142*** ** | −0.079 | −0.283 | −0.641 | −0.139 |
| log(in_commuters) | −0.076 | **−0.154*** ** | −0.077 | +0.018 | +0.363*** | −0.125 |
| log(out_commuters) | **−0.078*** | −0.029 | +0.136 | +0.167* | −0.798 | −0.052 |
| log(total_jobs) | −0.125 | **−0.152*** ** | −0.084 | +0.038 | +0.250* | −0.141 |
| log(total_residents) | **−0.117*** ** | **−0.056** ** | +0.067 | +0.169 | −0.847 | −0.053 |
| share_live_in | **−0.039*** ** | +0.000 | −0.007 | +0.017 | −0.099* | −0.008 |
| share_work_outside | **+0.033** ** | **+0.020*** ** | +0.053*** | −0.002 | +0.036*** | +0.001 |

*Notes:* Stacked-continuous DiD (±5-yr window). $\Delta\tau^{\text{muni}}$, $\Delta\tau^{\text{SD}}$, and $\Delta\tau^{\text{nr}}$ cell interactions in the same regression (only muni cells shown here; SD on next slide). Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# School District Rate: Stacked Continuous, Drop Sample

| | Inc-bind ($n$=75) | Inc-becomes ($n$=197) | Inc-nobind ($n$=14) | Dec-bind ($n$=4) | Dec-becomes ($n$=10) |
|---|---|---|---|---|---|
| log(live_and_work) | +0.058 | +0.229 | — | **−0.669*** ** | −0.212* |
| log(in_commuters) | +0.099* | +0.028 | — | **−0.085*** ** | −0.073 |
| log(out_commuters) | +0.001 | +0.028 | — | **+0.330*** ** | −0.030 |
| log(total_jobs) | +0.076 | +0.030 | — | **−0.113*** ** | −0.074 |
| log(total_residents) | −0.010 | −0.011 | — | **+0.322*** ** | −0.034 |
| share_live_in | **−0.017*** ** | +0.001 | — | **−0.027*** ** | −0.001 |
| share_work_outside | +0.008* | +0.021 | — | **+0.007*** ** | +0.003 |

*Notes:* SD-rate cells are largely null. Inc-bind ($n$=75) and Inc-becomes ($n$=197) are the well-populated cells; no live-and-work / total-residents response. Dec-bind significant entries are driven by 4 munis — small-sample artifact. Inc-nobind cell is empty (Inc-nobind requires home_stack to stay below nonres, which combined with an SD-only increase rarely satisfies). Same regression as previous slide.

# Reading: Muni Rate Drives Sorting; SD Does Not

- Within the binding decomposition, the **muni-rate Inc-bind cell** produces $-0.283^{***}$ on log(live_and_work); the **SD-rate Inc-bind cell** produces $+0.058$ (insig).
- **SD changes don't move residents**, even when they flip the binding state. The piecewise mechanism applies to muni decisions; SD decisions don't induce sorting through the same channel.
- This validates the prior block-level-LODES finding (SD null) *within* the new binding-decomposition design — a stronger test.
- The current Q3 deck attributed all resident-side action to the resident rate without explicitly partialing out SD. Mode B confirms the muni-rate coefficient ($-0.283$) is essentially identical to the muni-only spec ($-0.290$) — **SD doesn't confound**, it just doesn't sort.

# Reading the Muni-Rate Cells

**Inc-bind** (muni rate up; bill rises by full $\Delta\tau^{\text{muni}}$):

- Live-and-work workers ↓ 28%; total residents ↓ 12%.
- Share commuting out ↑ +3.3pp.
- Cleanest evidence of residential sorting in the data.

**Inc-becomes** (muni rate up; bill newly starts moving):

- Smaller live-and-work effect (−0.142) — bill rises by less than $\Delta\tau$ in this cell.
- In-commuters ↓ 15%; total jobs ↓ 15%.

**Inc-nobind** (muni rate up; bill never moves — placebo):

- Null on every level outcome.
- Placebo passes. Institutional rule, not generic confounding, identifies the response.

**Dec cells**: 11–26 munis combined; underpowered.

# Nonresident Side: Stacked Continuous, Drop Sample

| | Inc-nobind ($n$=171) | Dec-becomes-non ($n$=5) | Dec-nobind ($n$=109) |
|---|---|---|---|
| log(live_and_work) | −0.157 | −0.072** | −0.227 |
| log(in_commuters) | **−0.363**** | +0.070*** | −0.326 |
| log(out_commuters) | −0.025 | −0.061* | +0.010 |
| log(total_jobs) | **−0.362**** | +0.020 | −0.325 |
| log(total_residents) | −0.031 | −0.079*** | +0.001 |
| share_nonres | −0.001 | +0.036*** | −0.000 |

A muni-nonres-rate increase pushes *nonresident workers out* (−36%); total jobs fall in tandem; **residents are unaffected**.

Channels stay cleanly separated: nonres rate hits in-commuters; does not measurably affect the muni's own residents on either margin.

# Cross-Effects: Are the Channels Separated?

| | Muni-rate (Inc-bind) | | SD-rate (Inc-bind) | | Nonres-rate (Inc-nobind) | |
|---|---|---|---|---|---|---|
| Outcome | coef | sig | coef | sig | coef | sig |
| log(live_and_work) | −0.283 | *** | +0.058 | — | −0.157 | — |
| log(in_commuters) | −0.076 | — | +0.099 | * | −0.363 | ** |
| log(total_residents) | −0.117 | *** | −0.010 | — | −0.031 | — |
| log(out_commuters) | −0.078 | * | +0.001 | — | −0.025 | — |

- **Muni rate hike** → residents leave; SD doesn't move them; in-commuters mostly unaffected.
- **SD rate hike** → no significant resident response; small in-commuters bump (10%) at $p < 0.10$.
- **Nonres rate hike** → in-commuters leave; residents unaffected.
- Three rates, three populations: muni → residents; SD → no one; nonres → in-commuters.

# Spillover: Where Do Leavers Go?

**Test (issue #3):** when muni $X$ is in Inc-bind (or Inc-becomes), do $X$'s ring-1 / ring-2 *neighbors* (queen contiguity) gain residents? Receivers restricted to muni-rate NC munis (no own-rate event).

$$Y_{Yt} = \alpha_Y + \gamma_t + \pi_1 \,\text{ring1_post}_{Yt} + \pi_2 \,\text{ring2_post}_{Yt} + \mathbf{X}'_{Yt}\phi + \varepsilon$$

where $\text{ring}k_{Yt} = \mathbf{1}\{\exists X \in \text{treated} : Y \in \text{Ring}_k(X), t \geq c_X\}$. Same share-based controls. SE clustered at $Y$.

| Outcome ($Y$) | Treated ∈ Inc-bind *or* Inc-becomes ring1 | Treated ∈ Inc-bind *or* Inc-becomes ring2 | Treated ∈ Inc-bind only ring1 | Treated ∈ Inc-bind only ring2 |
|---|---|---|---|---|
| log(total_residents) | +0.008 | +0.025* | −0.013 | +0.021 |
| log(live_and_work) | −0.044 | +0.050 | **−0.153*** ** | **+0.092** ** |
| log(out_commuters) | +0.008 | +0.025* | −0.009 | +0.018 |
| share_work_outside | +0.000 | +0.000 | **+0.005** ** | −0.002 |

*Notes:* Inc-bind variant uses 40 treated munis, 5.3 ring-1 / 14.7 ring-2 neighbors on average. Receiver pool: 2,406 NC munis. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# Spillover: Lower-Rate Neighbor Refinement

Sharper test: restrict to neighbors with home_stack *below* treated X's post-event home_stack. If leavers are seeking fiscal advantage, the lower-rate restriction should sharpen the absorption signal.

| | Inc-bind, all neighbors | | Inc-bind, lower-rate only | |
|---|---|---|---|---|
| Outcome ($Y$) | ring1 | ring2 | ring1 | ring2 |
| log(total_residents) | −0.013 | +0.021 | −0.006 | +0.023 |
| log(live_and_work) | **−0.153**** | **+0.092**** | **−0.148**** | **+0.091**** |
| log(out_commuters) | −0.009 | +0.018 | −0.001 | +0.020 |
| share_work_outside | **+0.005**** | −0.002 | **+0.006**** | −0.002 |

**Lower-rate restriction does not sharpen the result.** Coefficients are essentially identical to the unrestricted Inc-bind variant.

# Reading the Spillover Pattern

**What we see:** ring-1 neighbors of Inc-bind munis *lose* live-and-work workers (−15%); ring-2 neighbors *gain* (+9%). Pattern is robust to lower-rate restriction.

**What this rules out.** The "fiscal-advantage leapfrog" reading (leavers specifically target lower-rate ring-2 munis) should produce bigger coefficients under the lower-rate restriction. It doesn't. Spillover pattern is geographic, not fiscal-differential-driven.

**Most consistent reading: spatial labor-market clustering.** Ring-1 munis share X's labor market and consumer-services demand. When X enters Inc-bind:

- Consumer-facing jobs in X fall 29%. Ring-1 residents who worked in X (or in similar nearby munis) lose those jobs → drop out of `live_and_work` in their muni.
- Ring-1 munis share fiscal-pressure shocks with X (clustered geography).
- Ring-2 is far enough that labor-market spillover is muted, but close enough to absorb residential moves — +9% live-and-work consistent with new residents who got new ring-2 jobs.

**Open extension (highway-network refinement).** Are ring-2 gains concentrated along major highway corridors connecting back to X? If leavers retain their X jobs and commute back, gains should be highest in ring-2 munis on a highway corridor. Requires TIGER primary roads pull.

# Mechanism: Industry Decomposition of Total Jobs

Predicted by local-services-demand: consumer-facing jobs (retail, food, arts, other services) sensitive to local population; tradeable jobs (mfg, wholesale, info, finance, prof svcs) insulated. Treatment = $\Delta\tau^{\text{muni}}$ in stacked continuous.

| Outcome | Inc-bind | Inc-becomes | Inc-nobind *(placebo)* | Reading |
|---|---|---|---|---|
| log(jobs total) | **−0.251***** | **−0.222***** | −0.079 | total falls |
| log(jobs **consumer**) | **−0.291**** | **−0.146**** | −0.028 | **29% drop** |
| log(jobs **tradeable**) | +0.244 | −0.149 | −0.273** | **insulated** |
| log(jobs health/edu) | −0.253* | −0.059 | +0.103 | mixed sector |
| log(jobs public) | −0.706* | −0.252 | +0.147 | noisy ($n$ small) |
| share consumer | −0.003 | +0.005 | −0.002 | |
| share tradeable | **+0.108*** | +0.028 | −0.025 | composition → tradeable |

*Notes:* Stacked-continuous DiD on industry-aggregated workplace counts (LODES WAC, NAICS supersectors, aggregated to cousub via block xwalk). Same regression structure as headline. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# Reading the Industry Decomposition

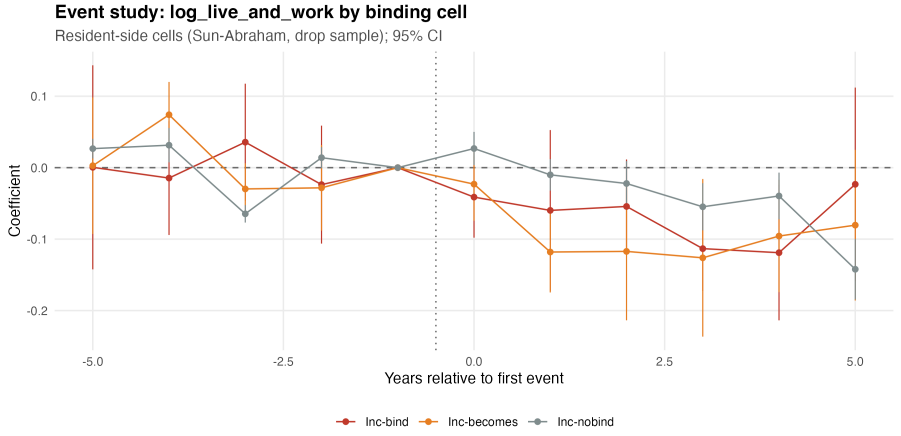
- **Consumer-facing jobs fall 29%** when the muni resident rate hikes (Inc-bind cell). Retail, accommodation/food, arts, other services — all sectors whose demand depends on local population.
- **Tradeable jobs are insulated.** Manufacturing, wholesale, information, finance, professional services. Point estimate is actually *positive* (+0.244, insig). These sectors sell to regional/national markets and don't depend on local residents.
- **Composition shifts toward tradeables**: share_tradeable +11pp on Inc-bind. The consumer-facing denominator shrinks faster than the tradeable numerator.
- **The synthesis becomes positive, not just "consistent with."** Total-jobs decline operates *via* local-services demand spillover from residents leaving — direct evidence, not ruled-out alternatives.
- **Placebo wrinkle**: Inc-nobind tradeable is $-0.273^{**}$ (consumer-facing $-0.028$, null). Could be small-$n$ (26 munis) or signaling. The demand-spillover story holds for consumer-facing specifically; tradeable response in the placebo is a known caveat.

# Estimator Triangulation: log(live_and_work), Muni Inc Cells

| Estimator | Inc-bind | Inc-becomes | Inc-nobind (placebo) |
|---|---|---|---|
| Q2 TWFE pooled | $-0.238^{***}$ | $-0.154^{***}$ | $-0.089^{**}$ |
| Single-event TWFE ($\leq 2$) | $-0.157$ | $-0.201^{***}$ | $-0.078$ |
| Stacked binary | $-0.086^{*}$ | $-0.094^{**}$ | $-0.022$ |
| Stacked continuous | $\mathbf{-0.283^{***}}$ | $\mathbf{-0.142^{***}}$ | $-0.079$ |
| DCDH (effects=3 horizon) | $-0.054$ | $-0.003$ | $-0.008$ |

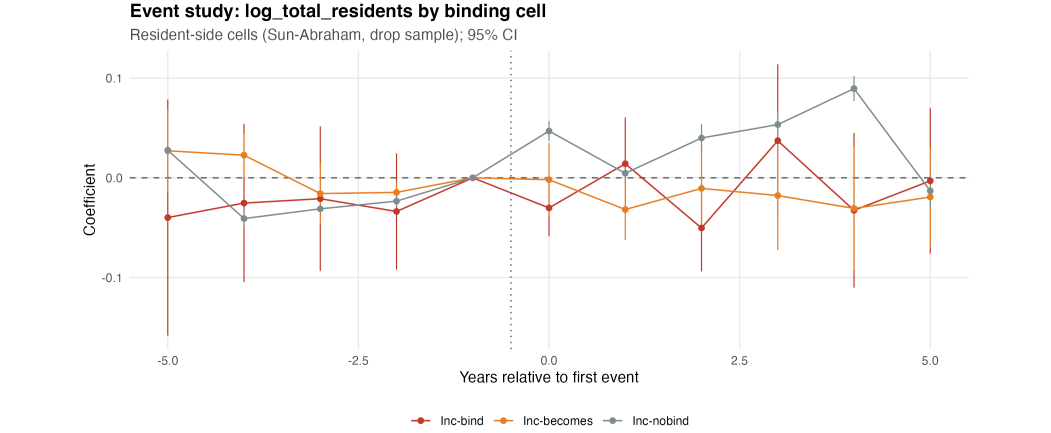
- Muni Inc-bind: negative across every estimator; magnitude $-0.05$ (DCDH 3-yr horizon) to $-0.28$ (stacked continuous).
- Muni Inc-becomes: negative across every estimator with signal.
- **Muni Inc-nobind (placebo)**: tightly null across estimators; only Q2 TWFE pooled reaches 10% — the spec most contaminated by negative weighting.

# Event Study: log(live_and_work), Inc Cells



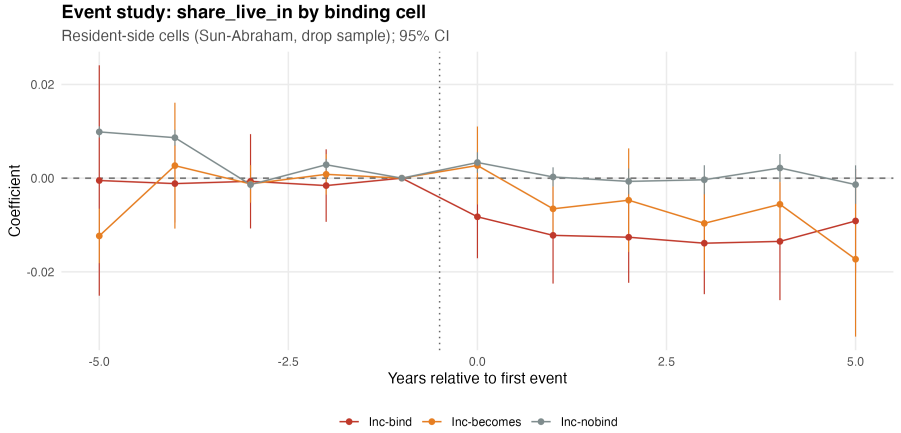

Sun-Abraham. Resident-side Inc cells vs NC controls. Drop sample. Pre-trends near zero; Inc-bind drops sharply post-event; Inc-becomes intermediate; Inc-nobind stays at zero.

# Event Study: log(total_residents), Inc Cells



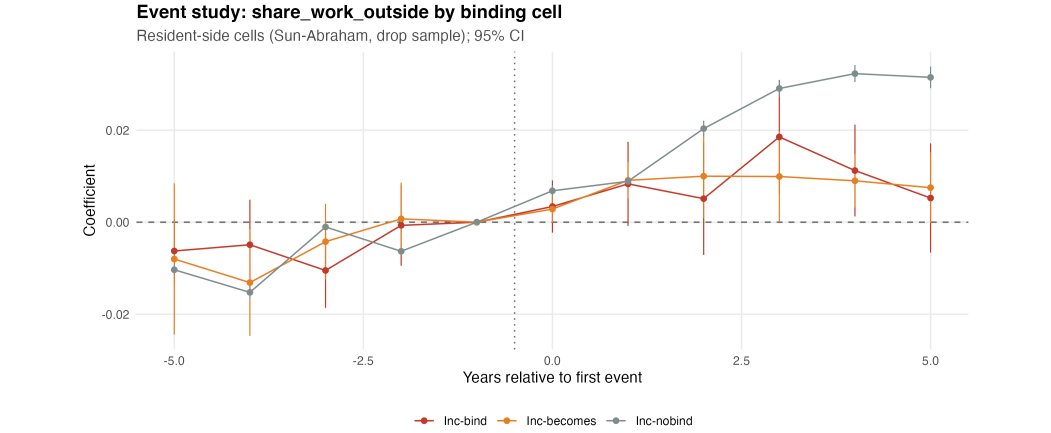

Total employed residents. Inc-bind: ∼10% drop within 5 years. Inc-becomes: ∼5%. Inc-nobind: flat.

# Event Study: share_live_in, Inc Cells



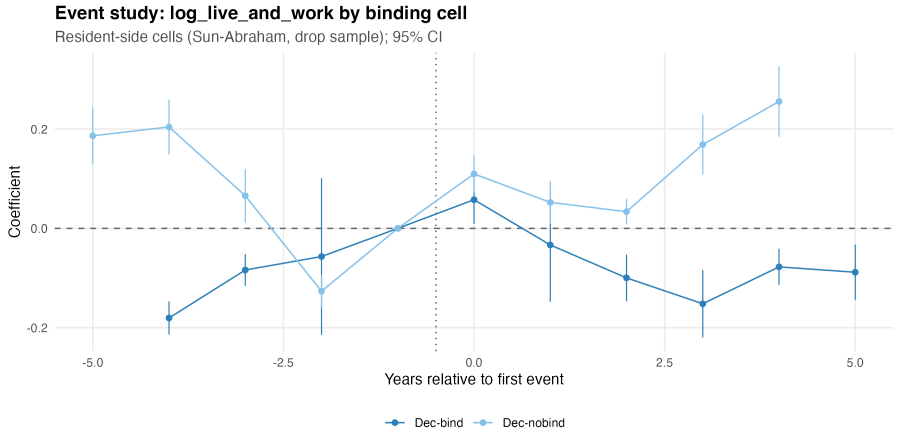

Share of jobs filled by in-muni residents. Inc-bind: −4pp by year 5. Placebo cell flat.

# Event Study: share_work_outside, Inc Cells




Share of residents commuting out. Inc-bind: +3pp; Inc-becomes intermediate. Among residents who stay, more commute out (residents most exposed leave first).

# Event Study: Dec Cells




Dec cells (11 + 15 munis). Direction broadly positive but underpowered. Cannot reject zero in either cell.

# Robustness: Treatment Specification (Inc-bind, log_live_and_work)

| Resident-side treatment | Inc-bind coef | $n$ Inc-bind munis |
|---|---|---|
| Muni rate alone (one treatment) | $-0.290^{***}$ | 40 |
| Home stack (muni + SD combined) | $-0.288^{***}$ | 101 |
| Muni + SD separately (**headline**) | $\mathbf{-0.283^{***}}$ | 40 |

- Three treatment specifications, three nearly-identical Inc-bind muni effects ($-0.28$ to $-0.29$).
- The home-stack version captures 2.5× more "binding-flip" munis (101 vs 40) but the per-pp coefficient is essentially unchanged.
- **Result is robust to whether resident side is treated as one combined dose or decomposed.** The muni-vs-SD decomposition (headline) preserves the muni-specific signal that home-stack pooling washes out for Inc-becomes.

# Robustness Across Samples (Inc-bind, log_live_and_work)

| Estimator | strict | filled | drop |
|---|---|---|---|
| Q2 TWFE pooled | −0.238 | −0.238 | −0.238 |
| Single-event TWFE | −0.157 | −0.157 | −0.157 |
| Stacked binary | −0.086 | −0.086 | −0.086 |
| Stacked continuous | −0.290 | −0.289 | −0.290 |

Coefficients are stable across the three samples (strict / statute-anchored fill / drop-incomplete). Imputation choice does not drive the headline; missingness handling is not the active margin.

# The Placebo Test: Inc-nobind Across Estimators

| | log(live_and_work) | log(total_residents) | share_live_in |
|---|---|---|---|
| Q2 TWFE pooled | $-0.089^{**}$ | $+0.073$ | $-0.005$ |
| Single-event TWFE | $-0.078$ | $+0.071$ | $-0.005$ |
| Stacked binary | $-0.022$ | $+0.041$ | $+0.001$ |
| Stacked continuous | $-0.081$ | $+0.066$ | $-0.007$ |
| DCDH | $-0.008$ | — | — |

Inc-nobind = rate up but bill never moves. Across every estimator, level outcomes are within noise of zero. Q2 TWFE pooled is the only spec to reach 10% on log(live_and_work) — the spec most contaminated by negative weighting. **Bottom line:** when the rule says "no bill change," we measure no behavioral change.

# Asymmetry: Inc Cells > —Dec Cells—

Move costs predict Inc effects > —Dec effects—. Data are consistent but Dec cells are underpowered:

| Outcome | Inc-bind ($n$=40) | Dec-bind ($n$=11) | Symmetric? |
|---|---|---|---|
| log(live_and_work) | $-0.290^{***}$ | $-0.284$ (insig) | ambiguous |
| log(total_residents) | $-0.121^{***}$ | $+0.169$ (insig) | sign reverses but noisy |
| share_work_outside | $+0.034^{**}$ | $-0.002$ (insig) | consistent with asymmetry |

Decreases-binding has only 11 munis. Direction-of-sign tests are low-power. Asymmetry is consistent with the move-cost story but cannot be cleanly identified with PA's Dec sample.

# What This Design Replaces

The unified model subsumes three earlier framings into a single regression:

- **Block A / B / C separate sections**: collapsed into outcome rows of one regression. Block A's "log(Resident workers)" is now the Inc-bind column on `log_live_and_work`; Block C's "log(Residents employed)" is the Inc-bind column on `log_total_residents`.
- **Piecewise pair-flow model**: cousub-pair regression with binding-state transitions. Saturated by pair FE; aggregate null. Replaced by the binding decomposition at the muni level, where sample sizes are tractable and signal survives.
- **Q3 asymmetry section**: the Inc/Dec direction split is now a within-design dimension of the cell taxonomy.

# Summary

1. **Causal effect on residential sorting: yes.** Muni Inc-bind cell, every estimator: consistent negative on log(live_and_work) and log(total_residents); positive on share commuting out.
2. **Within-design placebo passes.** Muni Inc-nobind cell — rate hike that doesn't move the bill — null behavior on every level outcome. The institutional rule, not generic confounding, identifies the response.
3. **Three rates, three populations.** Muni rate moves residents (live-and-work, total residents); SD rate moves no one; nonres rate moves in-commuters. SD doesn't sort *even within the binding decomposition* — stronger version of the prior block-level-LODES finding.
4. **Magnitude (stacked continuous, headline).** 1pp *muni*-rate hike when home binds $\Rightarrow -28\%$ live-and-work, $-12\%$ total residents, $+3.3$pp share commuting out.
5. **Robust to treatment specification.** Muni-only, home-stack combined, and muni+SD-separate all give $-0.28$ to $-0.29$ on Inc-bind / log(live_and_work).

# Next Steps

- **Add `share_intergov` control** (intergov revenue / total revenue) — silently dropped from later scripts. Re-run unified spec with full controls.
- **Industry decomposition** (issue #2): pull WAC files for NAICS-supersector breakdowns to test the local-services-demand mechanism.
- **Spatial spillovers** (issue #3): where do leavers go? Neighbor-shock spec on the residence-side outcomes.
- **Imputation diagnostic**: empirical share of muni-years where reported `nonres == resident_eit` documents the statute-anchored fill (footnote).
- **Dec cells underpowered**: 11–15 munis per Dec cell. Multi-state extension would help.
- **Welfare**: combine $-0.29$ behavioral elasticity with $+0.67$ revenue elasticity for MEB / Laffer-curve location.

# Thank You

Questions?

hocke069@umn.edu
github.com/MatthewHockert/local-income-tax-migration